iii. Reaction with alkalis: It reacts with strong caustic alkalis forming aluminates.

$$2\,Al + 2\,NaOH + 2\,H_2O \rightarrow 2\,NaAlO_2 + 3\,H_2\uparrow$$
(Sodium meta aluminate)

iv. Reaction with acids: With dilute and con. HCl it liberates H_2 gas.

$$2\,Al + 6\,HCl \rightarrow 2\,AlCl_3 + 3\,H_2\uparrow$$
(Aluminium chloride)

Aluminium liberates hydrogen on reaction with dilute sulphuric acid and liberates sulphur dioxide on reaction with hot concentrated sulphuric acid

$$2\,Al + 3\,H_2SO_4 \rightarrow Al_2(SO_4)_3 + 3\,H_2\uparrow$$
(Dilute)

$$2\,Al + 6\,H_2SO_4 \rightarrow Al_2(SO_4)_3 + 6\,H_2O + 3\,SO_2\uparrow$$
(Con.)

More to Know

Dilute or concentrated nitric acid does not attack aluminium, but it renders aluminium passive due to the formation of an oxide film on its surface.

v. As reducing agent: Aluminium is a powerful reducing agent. When a mixture of aluminium powder and iron oxide is ignited, the latter is reduced to metal. This process is known as **aluminothermic process**.

$$Fe_2O_3 + 2\,Al \rightarrow 2\,Fe + Al_2O_3 + \text{Heat}.$$

Uses

Aluminium is used in

- household utensils
- electrical cable industry
- making aeroplanes and other industrial mechine parts

8.8 METALLURGY OF COPPER

Occurrence:

It was named as cuprum by the Romans because they got it from the Island of Cyprus. Copper is found in the native state as well as combined state.

Ores of copper	Formula
Copper pyrites	$CuFeS_2$
Cuprite or ruby copper	Cu_2O
Copper glance	Cu_2S

The chief ore of copper is copper pyrite. It yields nearly 76% of the world production of copper. Extraction of copper from copper pyrites involves the following steps

i. Concentration of ore: The ore is crushed and the concentrated by froth floatation process.

ii. Roasting: The concentrated ore is roasted in excess of air. During the process of roasting, the moisture and volatile impurities are removed. Sulphur, phosphorus, arsenic and antimony are removed as oxides. Copper pyrite is partly converted into sulphides of copper and iron.

$$2\,CuFeS_2 + O_2 \longrightarrow Cu_2S + 2\,FeS + SO_2\uparrow$$

iii. Smelting: The roasted ore is mixed with powdered coke and sand and is heated in a blast furnace to obtain matte (Cu_2S + FeS) and slag. The slag is removed as waste.

iv. Bessemerisation: The molten matte is transferred to Bessemer converter in order to obtain blister copper. Ferrous sulphide from matte is oxidized to ferrous oxide, which is removed as slag using silica.

$$2\,FeS + 3\,O_2 \longrightarrow 2\,FeO + 2\,SO_2\uparrow$$

$$FeO + SiO_2 \longrightarrow FeSiO_3\ (\text{slag})$$
(Iron silicate)

$$2\,Cu_2S + 3O_2 \longrightarrow 2\,Cu_2O + 2\,SO_2\uparrow$$

$$2\,Cu_2O + Cu_2S \longrightarrow 6\,Cu + SO_2\uparrow$$
(Blister copper)

v. Refining: Blister copper contains 98% of pure copper and 2% of impurities and is purified by **electrolytic refining**. This method is used to get metal of a high degree of purity. For electrolytic refining of copper, we use:

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